IT1030 - Mathematics for Computing

Lecture 1
Number Systems

Number Systems

- Mathematical notation/symbols for representing values (numbers).
- In different systems, different symbols are used.

0123456789

٠١٢٣٤٥٦٧٨٩

I II III IV V VI VII VIII IX X

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Positional Number System

- A system for representation of numbers by an ordered set of numeral symbols (call digits).
- The value (or magnitude) of a numeral symbol (digit) depends on its position.
- A positional number system uses a **base** (or **radix**).
- A number system with base *b* is a system that uses distinct symbols for *b* digits.

Positional Number System (cont'd.)

- The 4 types of positional number systems are,
  - Decimal (Base = 10, {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}).
  - Binary (Base = 2, {0, 1}).
  - Octal (Base = 8, {0, 1, 2, 3, 4, 5, 6, 7}).
  - Hexadecimal (Base = 16, {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F}).
- We are used to dealing with numbers in the **decimal** system
- The base can be written as a subscript to the number for easy identification.
  - Example: 1265_{10} & 010000001_2

Positional Number System (cont'd.)

- To determine the quantity, it is necessary to multiply each digit by an integer power of the base and then form the sum of all weighted digits.

Example (7241_{10}):-

7		2		4		1	
×		×		×		×	
10^3		10^2		10^1		10^0	
=		=		=		=	
7000	+	200	+	40	+	1	
			7241				

Binary Number System

- Mainly used in computers and computer-based devices.
- A computer contains electronic components that uses voltages.
- Therefore, numbers are represented in a computer with a base of 2.
- Some other applications of binary number system are,
  - Microprocessor instruction
  - Graphics/Video
  - Sound

Number Base Conversion

- Should be able to convert a number in one base to another base.
- Examples will be discussed in converting,
  - Decimal ↔ Binary
  - Decimal ↔ Octal
  - Decimal ↔ Hexadecimal
  - Binary ↔ Octal

Repeated Division Method

- Can be easily used to convert a decimal number to another base.

1. Divide the number successively by the base.
2. After each division record the remainder.
3. The result is read from the last remainder upwards.

Repeated Division Method
(Example)

- Convert 123_{10} to a binary representation.
  - 123/2 ⇒ q = 61 & r = 1
  - 61/2 ⇒ q = 30 & r = 1
  - 30/2 ⇒ q = 15 & r = 0
  - 15/2 ⇒ q = 7 & r = 1
  - 7/2 ⇒ q = 3 & r = 1
  - 3/2 ⇒ q = 1 & r = 1
  - 1/2 ⇒ q = 0 & r = 1
- $123_{10} = 1111011_2$

Repeated Division Method
(Exercises)

- Convert below decimals to a binary representation.
  - 112_{10}
  - 673_{10}
  - 299_{10}
  - 1567_{10}

Repeated Subtraction Method

- Can be used to convert a decimal number to binary.

1. Starting with the 1st place, write down all of the binary place values in order until you get to the first binary place value that is GREATER THAN the decimal number you are trying to convert.

1. Mark out the largest place value (it just tells us how many place values we need).

Repeated Subtraction Method (cont'd.)

3. Subtract the largest place value from the decimal number. Place a “1” under that place value.

4. For the rest of the place values, try to subtract each one from the previous result.
   I. If you can, place a “1” under that place value.
   II. If you can’t, place a “0” under that place value.

5. Repeat Step 4 until all of the place values have been processed.

Repeated Subtraction Method
(Example)

$892_{10} = 1101111100_2$:

$2^9 = 512$	$892 - 512 = 380$			$\Rightarrow$	1
$2^8 = 256$	$380 - 256 = 124$			$\Rightarrow$	1
$2^7 = 128$				$\Rightarrow$	0
$2^6 = 64$	$124 - 64$	$=$	60	$\Rightarrow$	1
$2^5 = 32$	$60 - 32$	$=$	28	$\Rightarrow$	1
$2^4 = 16$	$28 - 16$	$=$	12	$\Rightarrow$	1
$2^3 = 8$	$12 - 8$	$=$	4	$\Rightarrow$	1
$2^2 = 4$	$4 - 4$	$=$	0	$\Rightarrow$	1
$2^1 = 2$				$\Rightarrow$	0
$2^0 = 1$				$\Rightarrow$	0

Repeated Subtraction Method
(Exercises)

- Convert below decimal to binary using the repeated subtraction method.
  - 33_{10}
  - 71_{10}
  - 123_{10}
  - 196_{10}

Other Base Conversions (cont'd.)

- Binary/Octal/Hexadecimal to Decimal

1. Find the position of every digit. We should count the position from the right direction of the number. And the position count starts from 0.
2. Multiply every digit with base to the power of their corresponding position. (base ^{position})
3. Finally, calculate the sum of all the multiples.

Other Base Conversions
(Examples)

- Binary to decimal

1111_2

$$= 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$= 8 + 4 + 2 + 1$$

$$= (15)_{10}$$

1001_2

$$= 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 8 + 0 + 0 + 1$$

$$= (9)_{10}$$

- Convert below binary to decimal representation.
  - 1101001_2
  - 101101_2
  - 10101_2
  - 110101001_2

Other Base Conversions
(Examples)

- Octal to decimal

16_8

$= 1 \times 8^1 + 6 \times 8^0$

$= 8 + 6$

$= (14)_{10}$

7777_8

$= 7 \times 8^3 + 7 \times 8^2 + 7 \times 8^1 + 7 \times 8^0$

$= 7 \times 512 + 7 \times 64 + 7 \times 8 + 7 \times 1$

$= 3584 + 448 + 56 + 7$

$= (4095)_{10}$

- Convert below Octal to decimal representation.
  - 140_8
  - 304_8
  - 1534_8
  - 110101001_2

Other Base Conversions
(Examples)

- Hexadecimal to Decimal

$25_{16} = 2 \times 16^1 + 5 \times 16^0$

$= 2 \times 16 + 5 \times 1$

$= 32 + 5$

$= 37$

$5BC_{16}$

$= (5 \times 16^2 + 11 \times 16^1 + 12 \times 16^0)$

$= 5 \times 256 + 11 \times 16 + 12 \times 1$

$= (1468)_{10}$

- Convert below Hexadecimal to decimal representation.
  - $7DE_{16}$
  - $E5F_{16}$
  - $9B4A_{16}$

Other Base Conversions (cont'd.)

- Binary to Octal

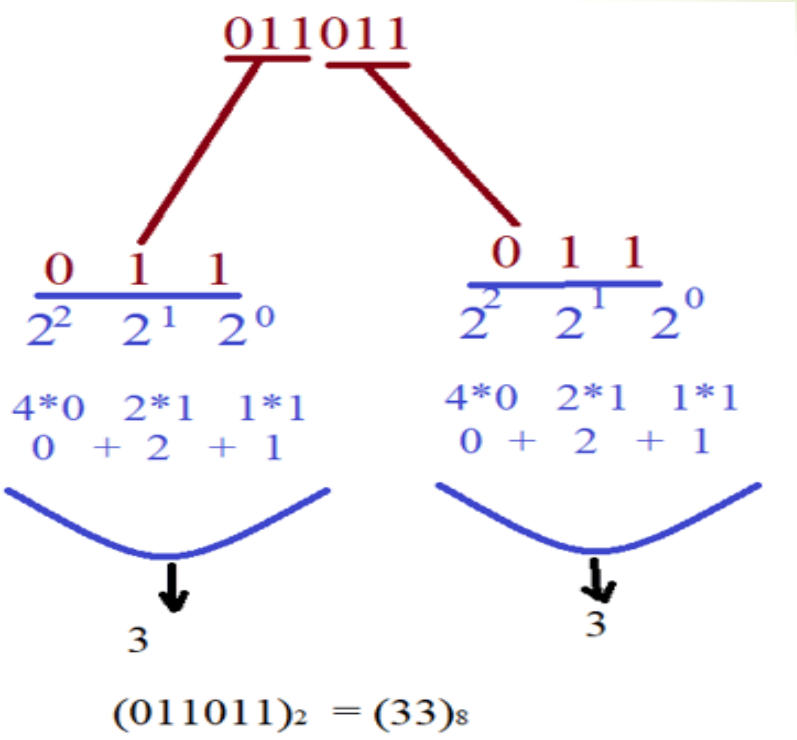
1. Form the bits into groups of **three** starting at the right and move leftwards.
2. Add 0's to the left if it does not form a group of three. Each group must have three digits.
3. Replace each group of three bits with the corresponding octal digit.

Other Base Conversions
(Examples)

- Binary to Octal

Exercise: Convert below given binary to octal

- 1010101_2
- 110011_2
- 11100111_2



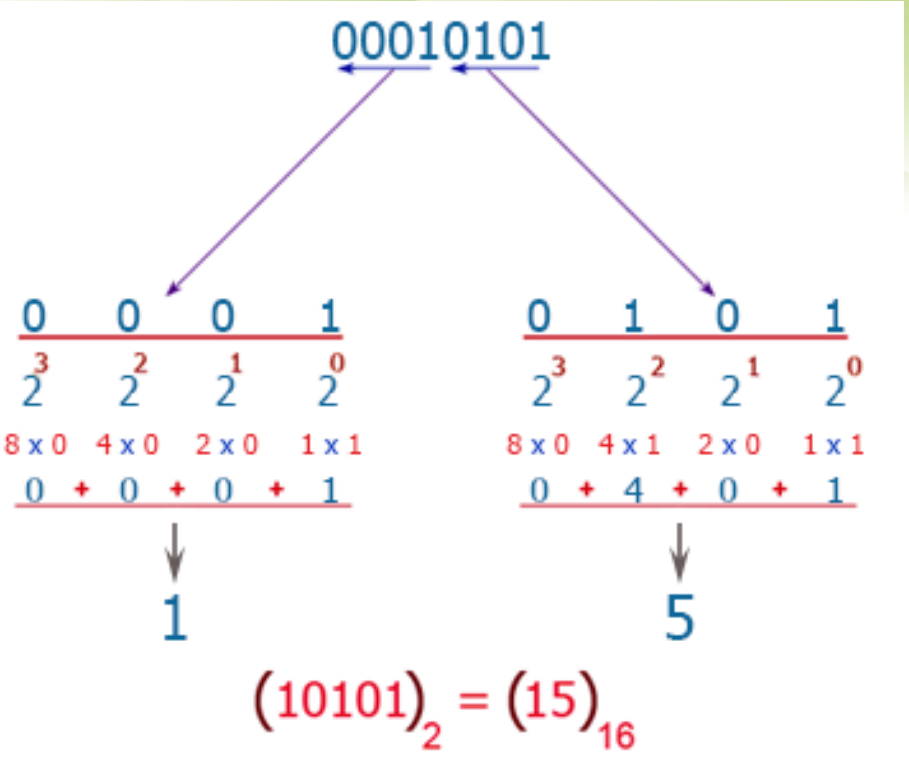
Other Base Conversions (cont'd.)

- Binary to Hexadecimal

1. Form the bits into groups of **four** starting at the right and move leftwards.
2. Add 0's to the left if it does not form a group of four. Each group must have four digits.
3. Replace each group of four bits with the corresponding hexadecimal digit.

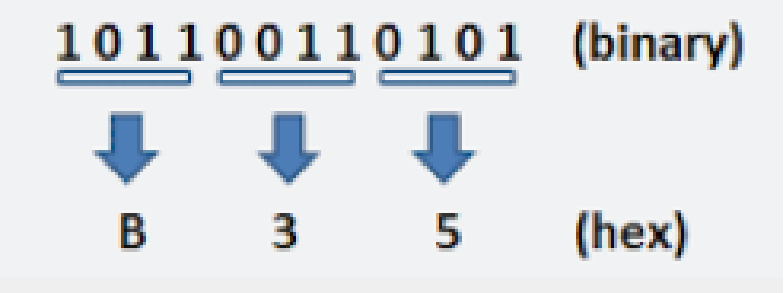
Other Base Conversions
(Examples)

- Binary to Hexadecimal



Exercise: Convert below given binary to hexadecimal

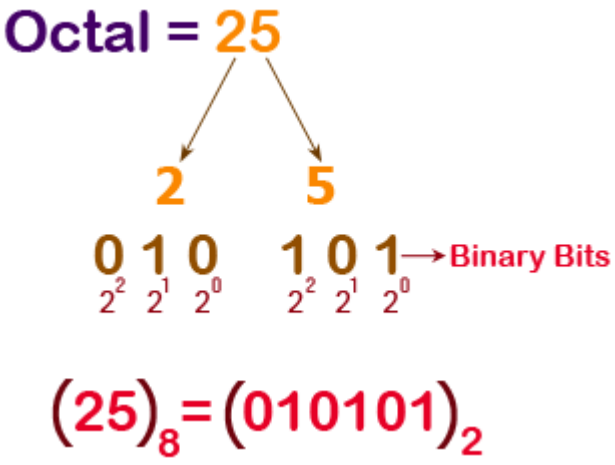
- 1010101_2
- 110011_2
- 11100111_2

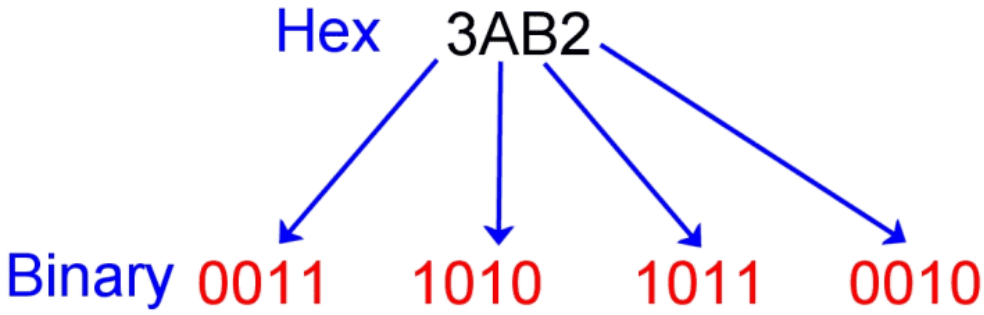


Other Base Conversions (cont'd.)

- Octal/Hexadecimal to Binary

1. The opposite of the above process is used.

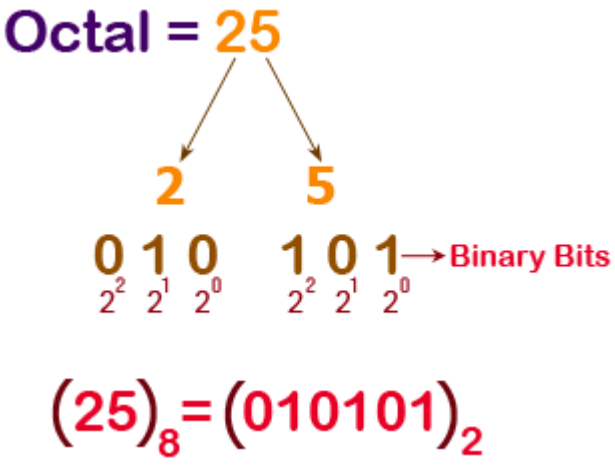


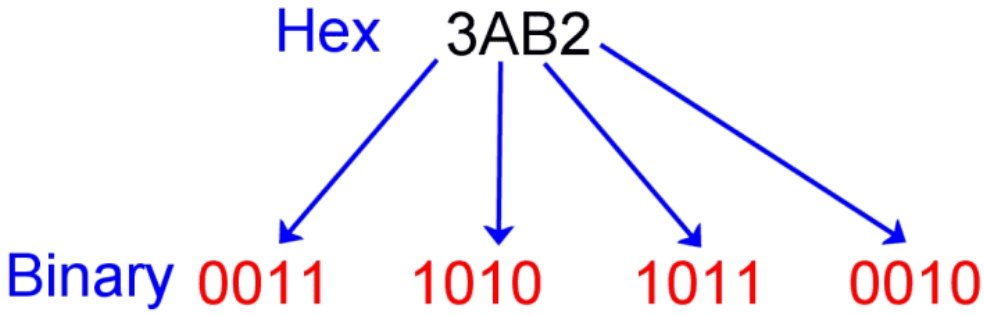


Other Base Conversions (cont'd.)

- Octal/Hexadecimal to Binary

1. The opposite of the above process is used.





Other Base Conversions (cont'd.)

- Convert below Octal to Binary
  - 56_8
  - 540_8
  - 4321_8
- Convert below Hexadecimal to Binary
  - $BC21_{16}$
  - $A2B_{16}$
  - FDA_{16}
  - 100_{16}

Conversion of Fractions

- Decimal Fractions to Binary Fractions

Convert the fractional part of decimal to binary equivalent

1. Begin with the decimal fraction and multiply by 2. The whole number part of the result is the first binary digit to the right of the point.
2. Disregard the whole number part of the previous result and multiply by 2 once again. The whole number part of this new result is the second binary digit to the right of the point.
3. Continue this process until we get a zero as our decimal part or until we recognize an infinite repeating pattern.

Conversion of Fractions
(Example)

- Decimal Fractions to Binary Fractions

1. Convert 0.125_{10} to binary.

$$(0.125)_{10} = (?)_2$$
$$= (0.001)_2$$

	Multiply by 2	Integer part
0.125	2	
0.25	2	0
0.5	2	0
1.0	2	1

Conversion of Fractions
(Example)

- Decimal Fractions to Binary Fractions

1. Convert 98.46_{10} to binary.

Binary equivalent of 98 is 1100010_2

Fractional part:

$0.46 \times 2 = 0.92 = 0$

$0.92 \times 2 = 1.84 = 1$

$0.84 \times 2 = 1.68 = 1$

$0.68 \times 2 = 1.36 = 1$

$0.36 \times 2 = 0.72 = 0$



2	98	
2	49	– 0
2	24	– 1
2	12	– 0
2	6	– 0
2	3	– 0
	1	– 1

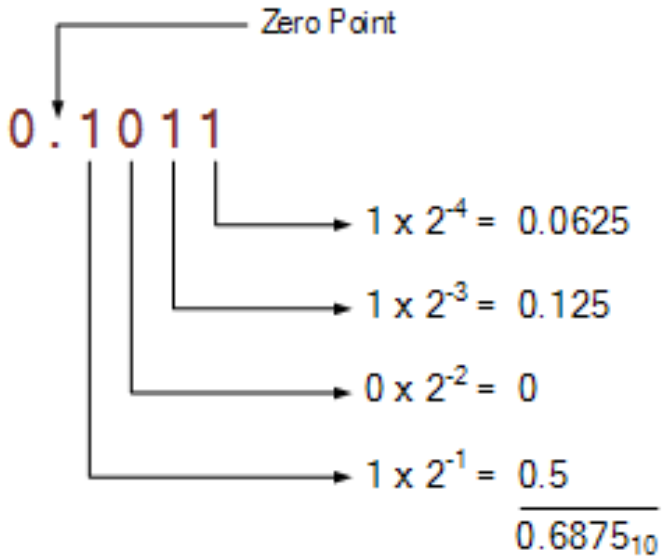
$1100010.01110\ldots_2$

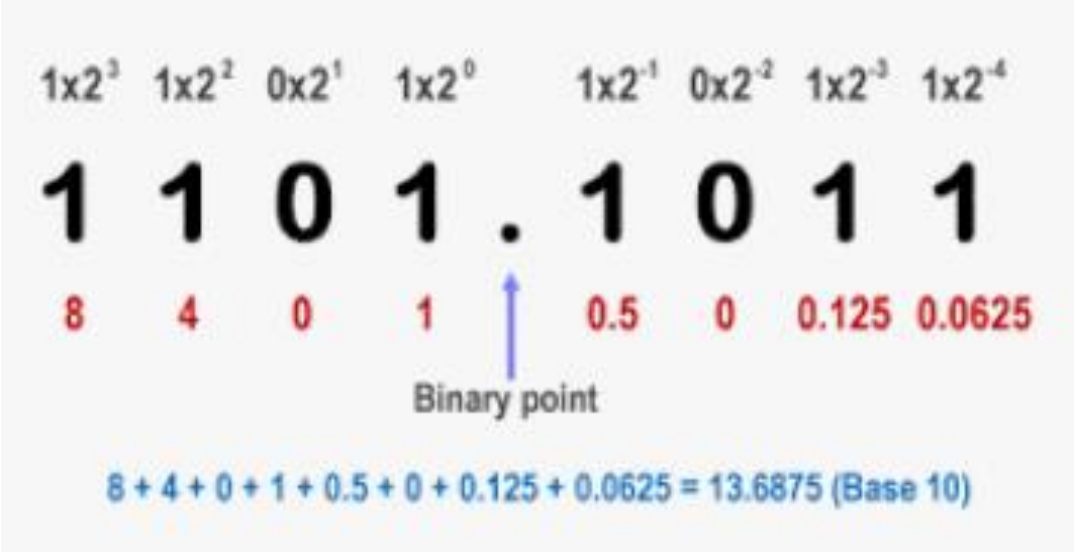
Conversion of Fractions
(Exercises)

- Convert below decimal fractions to binary fractions
  - 0.625_{10}
  - 0.1_{10}
  - 1.625_{10}
  - 6.986_{10}
  - 10.16_{10}
  - 1234.567_{10}

Conversion of Fractions (cont'd.)

- Binary Fractions to Decimal Fractions





Conversion of Fractions
(Exercises)

- Convert below binary fractions to decimal fractions
  - 10.01101_2
  - 110.101_2
  - 101.1101_2
  - 111.111_2
  - 1101.0111_2